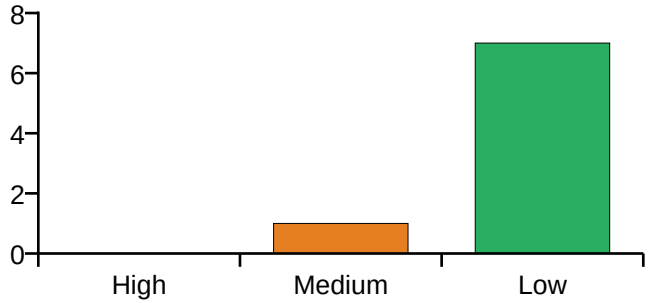
Defect Prediction Report

Report	-	**Generated**	2026-11-04 07:52 UTC
Source	data\samples\sample_project	**Submitted by**	CLI
Files analysed	8	**Units scored**	28 (23 functions, 5 classes)
Function model	v1 Random Forest / SMOTE (in-fold), CV AUC 0.734	**Class model**	v1 Random Forest / SMOTE (in-fold), CV AUC 0.741
Risk thresholds	Medium ≥ 0.4, High ≥ 0.7	**Skipped files**	0

Risk distribution



High: 0 file(s)

Medium: 1 file(s)

Low: 7 file(s)

Files ranked by defect probability

#	File	Lang	Risk	Prob.	Driving unit	Top contributing metrics
1	legacy/report_builder.py	python	Medium	0.489	build_monthly_report (line 11)	Total LOC=116 (+0.054) Distinct operators (n1)=35 (-0.022) Blank lines=11 (-0.020)
2	inventory/pricing.py	python	Low	0.218	compute_discount (line 5)	Blank lines=0 (-0.091) Executable LOC=40 (-0.037) Distinct operators (n1)=28 (-0.033)
3	java/OrderService.java	java	Low	0.196	OrderService::checkout (line 10)	Blank lines=0 (-0.121) Total LOC=15 (-0.046) Executable LOC=14 (-0.024)
4	inventory/models.py	python	Low	0.158	Product (line 17)	Class LOC=43 (-0.081) Public methods (NPM)=3 (-0.069) Weighted methods per class (WMC)=3 (-0.060)

#	File	Lang	Risk	Prob.	Driving unit	Top contributing metrics
5	scripts/load_fixtures.py	python	Low	0.135	<module> (line 1)	Total LOC=13 (-0.081) Blank lines=2 (-0.058) Executable LOC=10 (-0.035)
6	web/cart.js	javascript	Low	0.104	cartTotal (line 2)	Total LOC=9 (-0.100) Blank lines=0 (-0.046) Halstead content (I)=10.352 (-0.035)
7	inventory/utils.py	python	Low	0.076	chunks (line 13)	Total LOC=3 (-0.173) Blank lines=0 (-0.064) Executable LOC=3 (-0.036)
8	native/parser.c	c	Low	0.076	parse_int (line 5)	Total LOC=16 (-0.101) Blank lines=0 (-0.083) Executable LOC=16 (-0.042)

Notes

Non-Python files have partial metrics (no OO metrics; Halstead approximated).

Method: per-unit probabilities from Random-Forest-style models trained on NASA MDP (function level) and PROMISE CK (class level); a file's probability is the maximum over its units. Contributions are SHAP values (positive = raises risk). Models were trained on C/Java projects, so treat scores on other code as relative rankings.